Low-Frequency RoPE Subspace for Near–Far Sparse Attention

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***Abstract*—Sparse attention must retain useful candidates at low selection cost. We study a near–far page-to-token cascade sharing 32 low-frequency RoPE scoring coordinates and using original-dimensional attention on selected tokens. Near–far coverage involves trade-offs, and far-page reservation need not preserve far-token capacity. Author-reported 11-task RULER 32K AVGs are 58.30 and 59.29 under two Qwen3-8B page/token allocations; a Qwen3-32B comparison reports 50.64 sparse versus 52.22 dense. These descriptive comparisons neither isolate mechanisms nor establish non-inferiority. Separate historical timings show slower sparse decoding. Establishing benefit for the compact 4/128-protection design still requires matched quality and preparation-inclusive complete-request cost.**

***Index Terms*—sparse attention, shared representation, regional retrieval, long-context inference**

I. Introduction

In dense long-context decoding, each new token attends to a growing history of cached keys and values. Sparse attention restricts final attention to selected tokens, but selecting them requires scoring, ranking and memory accesses. The design goal is to preserve task-relevant tokens within a fixed budget while keeping selection cheaper than the attention work it removes. Both candidate selection and final attention-set size are therefore central design problems.

Cascades such as AsyncTLS shortlist pages using inexpensive summaries before token refinement [4]. Mean pooling and sign-aware extrema—summaries our selector also uses—do not rank pages by their total attention weight. Following ClusterKV's token/page perspective [7], Figure 1(a) shows a single Qwen3 query where fewer missed mass-ranked reference pages coincide with more missed reference-page mass. Unequal regional allocations and realized token counts preclude a controlled comparison of scoring rules or task quality. This motivates examining how scoring choices and page/token budgets jointly shape candidate access.

Candidate access also depends on how capacity is divided between recent and distant history. A global TopK can devote all available slots to one region, whereas regional quotas can reserve nominal capacity for each. In the diagnostic that selects whole bins in Figure 1(b), separate near and far reservations raise the mean retained fraction of far-region attention weight from .6759 to .7207, but reduce it in 43 of 128 correlated observations. The design must address both stages: admitting far pages alone does not reserve final slots for their tokens.

We study a two-level selector that couples a shared scoring view with explicit regional budgets. Both stages use a fixed set of 32 coordinates comprising 16 complete low-frequency RoPE pairs; separate near and far page quotas control eligibility for token refinement. Among the three 32-coordinate subsets in Figure 1(c), low-frequency pairs retain .84 of full-dimensional reference mid-TopK mass on average, random coordinates .51, high-frequency pairs .24; the random subset does not preserve RoPE pairs. Regional token quotas then divide the remaining budget after prefix and recent protection; selected tokens use original-dimensional Q/K/V in one final attention operation. Compact coordinates narrow scoring dot products, but ordinary token scoring still spans full history before masking; fewer shortlisted pages alone do not remove that work. We also formulate a prefill-informed far gate motivated by the layer variation in Figure 1(d); it is disabled by default and requires separate evaluation.

We contribute a shared-coordinate regional selector, joint page/token capacity analysis, and separate evaluations of task quality, regional coverage and execution cost. On complete 11-task RULER32K sets, reported Qwen3-8B AVG is 0.99 points higher at $\beta = .0625$ than at .125, with quotas changing together. A separate 11-task RULER32K comparison on Qwen3-32B shows a 1.58-point sparse deficit relative to dense attention. These descriptive comparisons do not isolate either mechanism, and the archived sparse configuration decodes more slowly than dense attention at every measured length. Establishing a quality–latency benefit for the compact 4/128-protection design requires matched task quality and complete-request measurements, including preparation, maintenance and selection.

II. Related Work

Quest supplies sign-aware page-extremum bounds for far-page scoring, SparQ approximates attention from query-selected key coordinates, and HISA prunes the attention set hierarchically [1], [2], [3]. AsyncTLS already combines a page-to-token cascade with coordinate selection, quantization, per-query normalization followed by grouped-query-attention (GQA) probability aggregation, and final attention over original KV [4]. SparQ also protects a recent-token window, while HISA forces the first and last blocks into its coarse candidate set [2], [3]. Rather than proposing a new cascade, we study these existing components with complete low-frequency RoPE pairs shared across both stages and separate nominal near–far L1 page quotas, distinguishing page-level eligibility from final token capacity. These page quotas do not by themselves guarantee final far-token capacity; ordinary L2 still scores the full history before masking.

Reducing attention traffic also requires considering what the selector itself reads. Token-Picker uses conservative probability estimates from partial key-bit chunks to stop further key-chunk and value transfers for pruned tokens [12]. In contrast to pruning token transfers from partial key bits, our selector ranks candidates with a fixed coordinate subset under page and token budgets, keeping original-dimensional KV for final attention. KV compression and placement methods address complementary storage and movement constraints [5], [6], [7]. A smaller scoring view retains original-KV capacity but does not itself establish lower complete-request latency.

III. Near–Far Selection in a Shared Subspace

To control both selection cost and candidate access, the cascade uses two explicit mechanisms: a shared scoring view and regional page quotas. One 32-coordinate map supports both L1 page screening and L2 token refinement, separating low-dimensional selection from original-dimensional attention (Fig. 2). The cascade operates only during decoding; prefill remains dense. L1 assigns separate page quotas to the near and far regions, so pages in one region do not compete for the other region's allocation. In the ordinary L2 path, full-history scoring precedes the retained-page mask and regional token selection. The selected token indices are merged and deduplicated

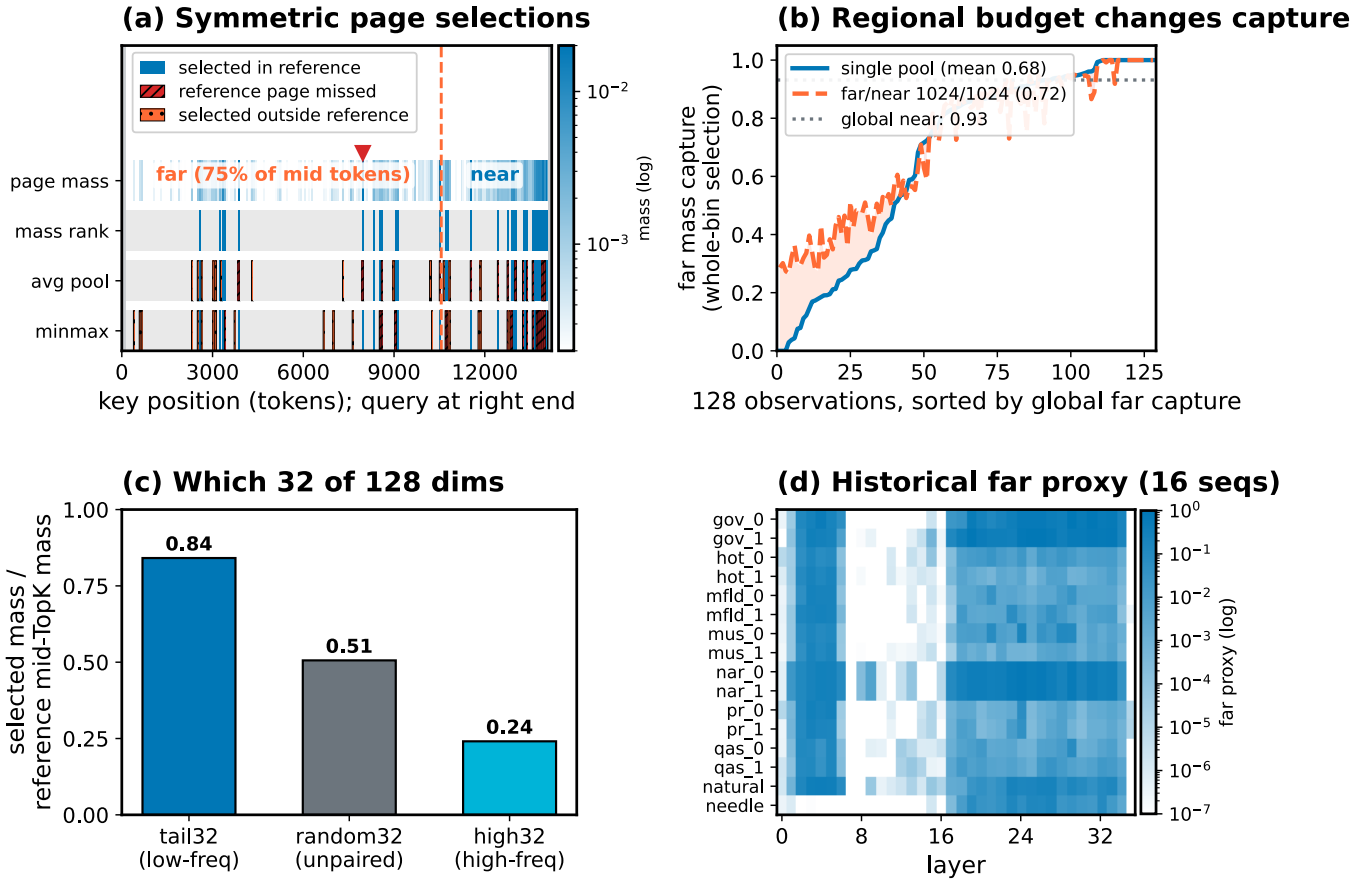


Fig. 1. Four motivating diagnostics; Qwen3-8B, 128/128 prefix/recent protection in (a)–(c). (a) MuSiQue sample 1, layer 24, query 14212. The top strip shows summed attention per page, not a selection. Below are the stored mass-rank, avg and minmax selections: 32 pages each, using 1989/1989/2048 tokens under a 2048-token cap. Mean/extrema miss 13/17 reference pages containing 59.65%/26.46% of reference-page attention mass, respectively. Blue denotes selected reference pages; red hatching denotes missed reference pages; orange dots denote selected pages outside the reference. Mass-rank is a whole-page comparison reference, not an optimal oracle, exact token TopK or DSA. Avg and minmax share 32 coordinates and head aggregation; the reference uses full-dimensional attention weights. Different regional allocations and realized token counts preclude inferring near/far specialization or task-quality ranking from these misses. (b) Four sequences contribute 128 correlated observations, ordered by global far retention. A global whole-bin cap of 2048 is compared with separate far/near caps of 1024/1024, without refill; red shading marks lower capture. Retention divides selected regional mass by full regional mass; the dotted line is global near retention. (c) Selected mass relative to full-dimensional reference mid-TopK mass; random32 does not preserve RoPE pairs. (d) One 16-by-36 historical single-query proxy matrix gives 24/36 layer medians below .01 and 12–32 low-proxy layers per sequence. It uses the last sampled query, group-logit summation before softmax and far interval $[64, t-2048)$ (S6), unlike the current gate. These are diagnostic selections and profiles, not task-quality or decode-safety results.

with protected positions for attention using the original Q, K and V; recomputing retained logits does not recover omitted tokens.

A. *Shared low-frequency coordinates*

Both regions and both selection stages use the same coordinate map within each layer and KV head, while each query head is scored separately. For Qwen3-8B, we select post-RoPE coordinates $J = \{48, \dots, 63, 112, \dots, 127\}$ from each 128-dimensional query and key. The indices retain 16 complete low-frequency split-half RoPE pairs (Supplementary Fig. S1). With coordinate-selection matrix $E_J \in \mathbb{R}^{128\times 32}$, the scoring vectors are $z_q = E_J^\top q$ and $z_i = E_J^\top k_i$. The retained pairs' slow positional rotation motivates this selection, although it does not guarantee preservation of important tokens. The compact implementation uses width-32 scoring tensors, whereas the historical quality runs used 32 active coordinates in width-128 L2 tensors (S7).

L1 aggregates selected key coordinates into page summaries for coarse ranking; L2 refines candidates using quantize–dequantize token keys. L2 uses 16 floating-point quantize–dequantize levels, not packed four-bit storage or INT4 arithmetic (S7). The two scoring views occupy separate buffers. Original KV remain full width. Compact scoring narrows each dot product fourfold, but view preparation and maintenance add work without compressing the original KV cache (S7).

B. *Regional page and token budgets*

In decoding, t indexes the current token from zero across the full sequence including prefill. Let s and w denote the protected prefix and recent-window lengths. Their union and the unprotected middle are

$$\mathcal{P}_t = \{i : 0 \le i \le t,\ i < s \text{ or } i > t - w\}, \tag{1}$$

$$\mathcal{M}_t = \{0, \dots, t\} \setminus \mathcal{P}_t. \tag{2}$$

Overlapping protection counts once. Near is a suffix of $\mathcal{M}_t$; far is the rest of $\mathcal{M}_t$. Near excludes the protected recent window. Near uses fraction α of unprotected history, a one-page minimum and a downward page-aligned start; S7 specifies integer rules.

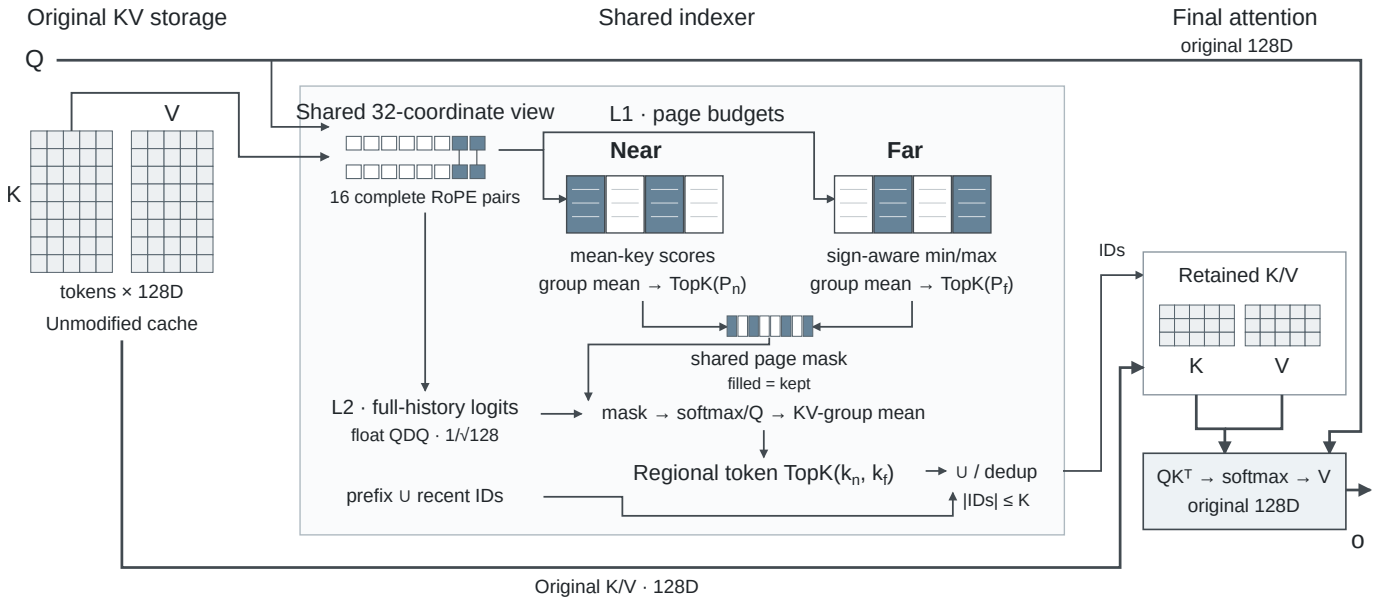


Fig. 2. Architecture of shared-subspace selection, with the optional gate inactive. A shared coordinate view feeds regional L1 page scoring and ordinary full-history L2 scoring. Page masks limit eligibility; per-query softmax and KV-group probability averaging precede regional token selection. Protected positions join deduplicated IDs within K. Near excludes the protected recent window. IDs define support over original-width K/V; physical gathering or dense masked computation depends on the backend. Original Q follows a separate path. Arrays are schematic.

Regional page caps and token quotas constrain the cascade (Supplementary Fig. S2). Near pages use query–mean-key dot products to summarize page-level relevance; far pages use sign-aware extrema to bound unquantized shared-space token scores. This regional assignment is a heuristic. Summaries cover full physical pages, including protected boundary positions; L2 separately limits token regions (S7). For a nonempty page b, let $m_{b,j} = \min_{i\in b} z_{i,j}$ and $M_{b,j} = \max_{i\in b} z_{i,j}$. Quest's construction [1] gives

$$U_b(z_q) = \sum_j (z_{q,j}^+ M_{b,j} + z_{q,j}^- m_{b,j}), \tag{3}$$

where $z^+ = \max(z, 0)$ and $z^- = \min(z, 0)$. Each signed coordinate contribution is bounded by its chosen extremum, so $z_q^\top z_i \le U_b(z_q)$ for every unquantized shared-space key in b. This does not bound original-dimensional logits or independently quantized scores. Extrema may come from different tokens, making the bound loose enough to misrank pages and miss important tokens. Average-key scores lack this per-token bound.

For total cap $K \ge |\mathcal{P}_t|$, define the middle-token budget $K_{\text{mid}} = K - |\mathcal{P}_t|$. α controls near extent, β its share of L1 page cap K_1, and γ its nominal L2 quota from retained-page capacity. For the partitioned fixed-quota path (far active; no overrides), assume integer $K_1 \ge 1$, $\alpha > 0$, $0 < \beta \le 1$ and $\gamma \ge 0$. With page size B and sufficient regional pages/tokens, $P_n = \max(1, \text{round}(\beta K_1))$, $P_f = \max(0, K_1 - P_n)$, $k_n = \min(\lfloor \gamma P_n B \rfloor, K_{\text{mid}})$ and $k_f = \max(0, K_{\text{mid}} - k_n)$; rounding uses ties-to-even. Raising β cannot reduce the nominal near-page quota or increase the far-page quota; rounding can leave both unchanged. Thus $P_f > 0$ does not imply $k_f > 0$: page and token access are separate constraints. For example, $K_1 = 128$ halves the near-page quota from 16 to 8 when β drops from .125 to .0625, hence far pages rise 112→120 under the unchanged cap. With enough positive-probability candidates, L2 cannot recover unprotected tokens from L1-rejected pages. S7 details scarce-support and masked zero-tie exceptions.

With disjoint protection and sufficient candidates, the compact configuration ($K = 1024$, $s = 4$, $w = 128$) leaves 892 middle slots per KV head. Here $\gamma = .625$ yields $16 \cdot 64 \cdot .625 = 640$ near and 252 far slots (Supplementary Fig. S2). The partial prefix page can reselect up to four protected tokens, so 252 is nominal rather than guaranteed unprotected far coverage.

C. GQA selection and original-dimensional attention

GQA assigns four of Qwen3's 32 query heads to each of its eight KV heads. L1 averages their page scores before regional TopK. Ordinary L2 computes each query's full-history logits, applies the shared page mask, then softmax. Group-averaged probabilities determine regional token TopK. L2 uses scale $1/\sqrt{128}$ before softmax even with 32 scoring coordinates. Because softmax precedes group averaging, this rule generally differs from applying softmax to group-averaged logits. Width-32 scoring narrows each dot product; a smaller L1 shortlist restricts eligibility but does not reduce ordinary L2's full-history dot-product count.

For KV group h, let $\mathcal{S}_{t,h}$ be the deduplicated union of protected positions and regional winners, with $|\mathcal{S}_{t,h}| \le K$. Each query g uses this support for its own full-dimensional attention ($d = 128$):

$$o_{h,g} = \sum_{i\in\mathcal{S}_{t,h}} \frac{e^{q_{h,g}^\top k_{i,h}/\sqrt{d}}}{\sum_{j\in\mathcal{S}_{t,h}} e^{q_{h,g}^\top k_{j,h}/\sqrt{d}}} v_{i,h}. \tag{4}$$

Each query applies one softmax over the merged indices, avoiding the addition of separately normalized regional outputs [8]. Full-dimensional retained logits still use a restricted denominator, which can change the output. These indices define a shared logical support. Physical KV gathering and reuse depend on backend and length; short sequences can use a dense masked path.

D. Prefill-informed far gating

Beyond scoring and budgeting, layer variation offers a second savings opportunity. The optional gate tests whether low far-region

TABLE I
Compact design parameters and execution policy. These settings define the proposed implementation, not the historical runs. K includes unique protected tokens.

Parameter	Setting
Final token cap K	1024
Page size; L1 page cap K_1	64; 128
α, β, γ	.25, .125, .625
L1/L2 active coordinates	32/32
Prefix / recent protection	4/128
Prefill; decode layers	dense; all sparse
L2 scoring support	full history before mask

attention during prefill can identify layers where far selection may be omitted during decoding; it is disabled by default. For each input and layer, a mean far-region attention proxy below ϵ sets the gate OFF; otherwise far selection stays active. Calibration uses a token-defined region that can differ from the page-aligned skipped region (S5). The first decode-path call freezes the decision until input reset (S5). OFF skips far L1/L2 scoring and selection in the supported fixed-quota path, refilling near L2 within unchanged near pages. Removing far support can change softmax denominators and group-averaged near rankings (S5). Representation preparation, sequence-wide score buffers and normalization remain.

The gate supports one sequence; the skip path requires valid length and nonempty far history (Supplementary Fig. S3; S5). The prefill threshold does not bound decoding error; gate quality and total cost require separate evaluation.

IV. Experimental Setting

We evaluate three aspects separately: task quality (Q1), regional candidate coverage (Q2) and execution cost (Q3). Table I defines the compact Qwen3-8B design: dense prefill, sparse decoding and a 1024-token cap including four prefix and 128 recent tokens (the 4/128-protection design). The historical five-task LongBench shared-subspace arm is linked to an earlier implementation: 128/128 protection and 32 active coordinates stored in zero-filled width-128 L2 keys. The comparison labels "full-dimensional page scoring" and "full-dimensional token scoring" denote reported variants whose complete execution configurations are unavailable. We retain these nominal labels without treating the runs as controlled single-stage ablations. Synthetic selector timing and model-level decode timing come from separate studies; none supplies matched quality and cost for the compact design. S6/S7 give configuration and measurement details.

Workloads and comparisons. Reported quality covers five-task LongBench [9] and 11-task RULER [10] (Qwen3-8B: 32K/64K, historical 128K; Qwen3-32B: 32K), plus ported Quest/AsyncTLS/JSSA baselines and a Llama-3.1-8B arm at 32K. Three nominal sparse LongBench variants lack verified factorial isolation. FullKV is dense and uncapped (RULER/decode timing). Native protocols retain disclosed prefill/decode; policy-matched adaptations need separate reporting. Sparse-decode budget matching aligns final unique-token caps K including protection, not merely labels or coarse coverage. Realized counts and head sharing affect KV traffic; page caps may differ. Equal K alone establishes neither quality nor complete-request cost matching (S7).

Metrics. Primary quality is task-score AVG over each fixed benchmark/length set. Per-task gains and losses explain AVG; zero loss on every task is not required. Policy-matched controls require matched models, samples, lengths, scorers and phase policies, and aligned sparse-token budgets. Incomplete sets lack full-set AVG. Regional retention in Fig. 1(b) divides selected mass by full regional mass; Fig. 1(c) instead uses full-dimensional reference mid-TopK mass. Selector time covers a synthetic single-layer operation; synchronized TPOT measures model-level intervals after the first output token. Neither measures complete-request latency, including preparation, maintenance and input/output work. Unpaired quality and timing studies do not define a quality–latency frontier. AVG differences are descriptive, not statistical non-inferiority evidence.

V. Evaluation and Discussion

A. Q1: Does reduced scoring width preserve task quality?

Five-task means sit within a descriptive 0.33-point spread across the three nominal scoring configurations. On HotpotQA, Qasper, MuSiQue, GovReport and RepoBench-P [9], shared-subspace scoring scores 47.588, the nominal full-dimensional page variant 47.910 and the token variant 47.666 (Supplementary Fig. S4), the shared-subspace arm lower by 0.322 and 0.078 points. These differences describe recorded configurations, not evidence that reduced scoring width preserves quality: the variants are not verified single-factor interventions, and the shared-subspace run uses earlier 128/128 protection. Neither the stages' representation effects nor the compact design's quality follows; all three arms are sparse, dense FullKV a different reference.

B. Q2: How do regional budgets constrain candidate access?

Regional quotas lower far retention in 43/128 correlated observations despite improving its mean (Fig. 1(b)); this is not task-quality evidence. Reported 8B 32K AVG rises 58.30→59.29 when $\beta = .125 \rightarrow .0625$ at $K_1 = 128$, $K = 1024$, protection 4/128. Pages 16/112→8/120 and tokens 640/252→320/572 change jointly. At $\beta = .03125$, AVG falls to 58.96 ($K_1 = 256$) and 58.07 ($K_1 = 128$, far tokens 732), bracketing the $\beta = .0625$ peak; CWE rises 3.9 and FWE falls 9.67. Data/scorer records match these sparse arms; predictions remain unreplayed.

Reported 32K AVG gaps to dense FullKV are -0.09 (8B, different scorers) and -1.58 (32B, matching manifests), across machines. At 64K, 8B AVGs are 48.85/48.05 ($\beta = .0625/.125$) versus 48.54. The sparse arms share a machine and scorer hashes but differ in prediction scripts; the separate-run dense anchor has a different scorer and legacy-partial identity. These mismatches and missing paired intervals preclude causal or non-inferiority conclusions (S8).

Budget-aligned comparisons. At AsyncTLS's token budget of 512, the half-budget arm scores 56.00 against the port's 50.09 ($+5.91$); at $K = 1024$, 59.29 versus JSSA's 58.7 (budget 1024) and Quest's 42.85; $K = 384$ collapses to 15.00. Migrating $\beta = .0625$ to Llama-3.1-8B lowers AVG 86.19→83.88. Port provenance, CWE's 43.4-point drop and missing paired intervals are in S8.

At the farthest length, a same-machine 128K pair retains 98.49% of the 46.43 dense anchor (45.73 at $\beta = .0625$, 46.02 at $\beta = .125$). Historical RULER 128K yields 23.98 (sparse) under budgets implying $k_f = 0$, execution unverified, the 22.45-point loss unexplained (S6). Five-task LongBench means are 26.09 (protection-only, 256 tokens) versus 47.91 (reused sparse, 1024); retrieval and budget both change.

C. Q3: Does compact scoring reduce execution cost?

Model-level decoding. The reported sparse configuration is slower than dense FullKV at every measured length (Fig. 4). Sparse TPOT medians are 87.9, 97.6 and 119.9 ms at 32K, 64K and 128K, compared with dense medians of 26.0, 39.7 and 67.3 ms. Although the relative penalty decreases with length, absolute sparse TPOT increases. Thus these measurements show no decode-time benefit

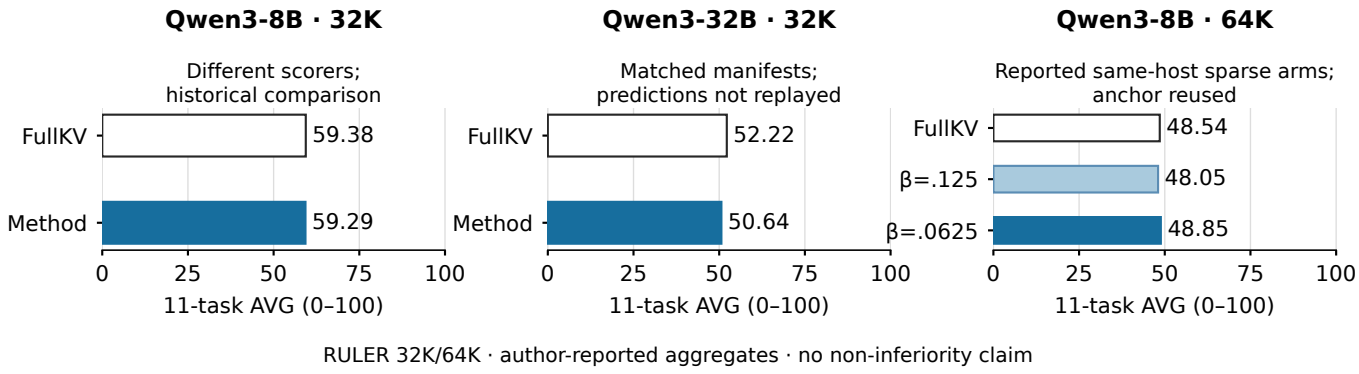


Fig. 3. Author-reported RULER 32K/64K AVG: 11 tasks, 100 examples/task/arm. FullKV is dense; sparse arms report $K = 1024$, protection 4/128. At 32K, $\beta = .0625$ (8B) or .125 (32B); at 64K, both β values are shown for 8B. The 32K comparisons span machines (8B scorers differ; 32B manifests match, predictions unreplayed); the 64K sparse arms are reported on one machine, the FullKV anchor a separate run. Gate status and scoring dimensions are unverified. These descriptive scores establish neither non-inferiority nor matched quality–cost benefit (S8).

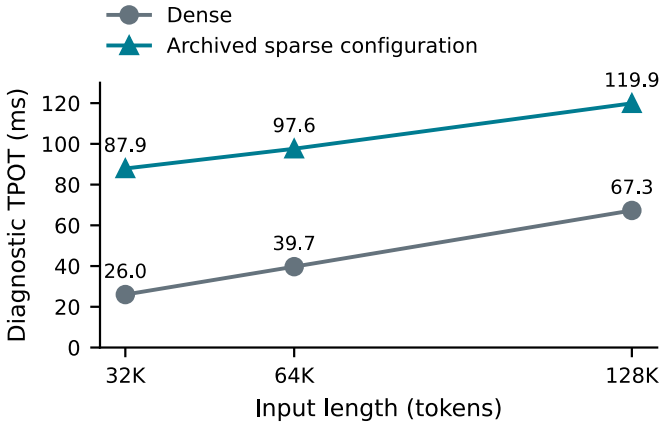


Fig. 4. Author-reported model-level decode TPOT for an archived sparse configuration and dense FullKV, Qwen3-8B with dynamic cache. Fixed 128-token output gives 127 after-first intervals, including device synchronization and argmax. The archived $(.125, .375, .625)$ setting differs from the selected 4/128 candidate. These medians do not measure complete-request E2E or establish a matched quality–cost comparison; provenance limits are in S6.

from that implementation. They include device synchronization and argmax, exclude tokenization, transfer and output, and use a configuration different from the compact 4/128 design. Aggregate medians cannot locate the cause or predict the compact implementation (S6).

Local selector cost. In the synthetic study, a separate two-round experiment (single layer, batch one, one query, 64K QKV) reports steady-state summaries of 1.3502 ms for the full-dimensional page variant, 1.3598 ms for shared-subspace scoring and 1.4634 ms for the token variant, so the shared-subspace aggregate time exceeds the page variant's. Shared-subspace scoring runs 0.0237 ms slower in round one and 0.0046 ms faster in round two, with no consistent advantage across rounds. Missing raw sequences and an unresolved sampling-protocol discrepancy prevent reliable assessment of these small differences or a timing-equivalence conclusion (S6). These local timings identify no bottleneck, lack LongBench quality pairing and measure no complete-request cost.

D. Optional gate: prefill is not a decode guarantee

Low prefill attention does not establish safe skipping of far selection during decoding. An archived input–layer record pairs a prefill far-attention proxy below .05 with a reported decode-tail mean of .1678 over the last five queries; a proxy shift, not measured task-quality loss. Gate thresholds .05 and .10 yield five-task AVGs of 47.876 and 47.692 against the reused 47.910 reference, down .034 and .218 points. While the AVG decreases are small, three tasks decline per arm, led by HotpotQA's 1.05 and 1.26; AVG remains primary, describing heterogeneity, not a per-task zero-loss requirement. Neither these AVG changes nor task-bootstrap intervals containing zero establish non-inferiority; missing scoring configurations, paired predictions and matched costs leave gated quality and net savings unverified (S6).

Design implications and limitations. Scoring and allocation rules expose two distinct constraints. Reducing scoring width narrows each dot product, but ordinary L2 still scores and normalizes the full history. Far-page quotas admit candidates to refinement without guaranteeing final far-token capacity. A useful operating point reduces complete-request latency at matched task quality, counting preparation, maintenance and selection once. The historical sparse configuration decodes more slowly than dense attention, and separate selector timings establish no complete-request latency benefit. Matched task-quality and complete-request measurements for the compact 4/128 design remain missing; historical gate summaries lack complete shared-subspace configuration identity.

VI. Conclusion

We studied shared low-frequency page/token scoring with regional allocation and original-dimensional attention on selected tokens. The analysis separates narrower scoring from reduced work and far-page reservation from far-token coverage. Qwen3-8B RULER 32K AVG is 0.99 points higher at $\beta = .0625$ than at $\beta = .125$ with quotas changing jointly; $\beta = .03125$ arms score 58.07/58.96, bracketing the peak. A Qwen3-32B sparse configuration trails dense attention by 1.58 points. A same-machine 128K pair retains 98.49% of the dense anchor; migrating $\beta = .0625$ to Llama-3.1-8B costs 2.31 points. These descriptive results neither isolate mechanisms nor establish non-inferiority. Historical 128K loss and slower sparse decoding remain configuration limitations. Establishing benefit for the compact 4/128 design still requires matched quality and complete-request cost.

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